

g·SENSOR

BIOSIGNAL SENSOR

INSTRUCTIONS FOR USE





Instructions for Use: g.Sensors

Version: 1.25.01

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www.gtec.at | office@gtec.at

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1 ABOUT THIS DOCUMENT

g.tec provides utilities and various sensors for physiological and physical signals. All sensors are designed to be connected to the input channels of g.tec amplifiers to enable synchronous and simultaneous recording of many electrophysiological data (including EEG, ECG, EMG, EOG and ECoG). This manual describes how to use different sensors in a correct and safe way to assure best data quality for research. The technical principles of the sensor functions are explained and recommendations for the amplifier configuration are given. The manual also shows the correct application of the sensors and provides procedures to test functions. Mandatory safety measures, technical data and specifications are listed.

1.1 IMPORTANT INFORMATION FOR USING A SENSOR OR UTILITY

Any user of our sensors, utilities and amplifiers is responsible for the correct application and safe operation. Before use, please carefully read these instructions and any documents that may be packed together with some sensors.



NOTE

g.tec's sensors and utilities are no certified medical devices and are not intended to be used as medical devices. Use g.tec's sensors and utilities for research applications only.

1.2 CLEANING AND STORAGE OF SENSORS AND UTILITIES

Unless otherwise stated in the manufacturer's documents (which may come packaged together with some sensors), please note the following instructions. Use only a damp cloth when cleaning devices, cables and connectors. Use alcohol (70%) with a soft tissue for disinfection purposes if required. Ensure that any liquids do not enter the device or connectors. Make sure that all components are completely dry before storing the device. If applicable, remove all batteries from the device before storage or if the device will not be used for at least 2 weeks. Store the device in a dry place between -10°C and +50°C. In case the device was stored at lower temperatures, wait until all condensed water disappears before using it (wait at least 1 hour in a heated room).

1.3 CONNECTING SENSORS TO AMPLIFIER INPUTS

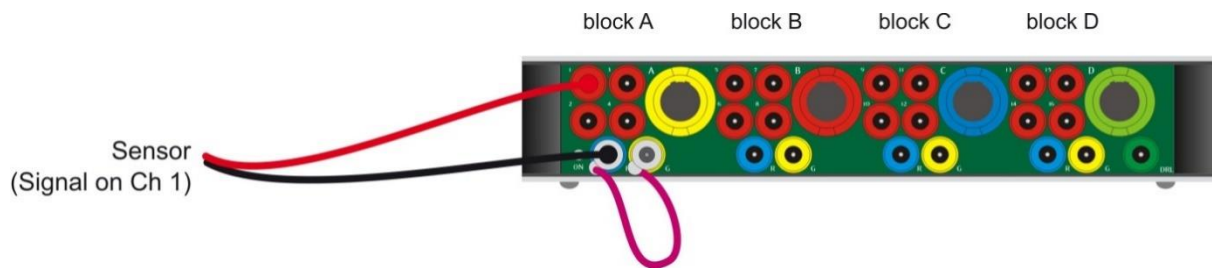
All sensors provided by g.tec can be connected directly to the input channels of our biosignal amplifiers. Make sure that sensors are not connected to any other electrical devices, power supplies or charging devices. Do not connect the output leads of a sensor to electrodes or to other electrical devices. Patient safety is provided by the biosignal amplifier's applied part or medical data acquisition system. Using a sensor connected to an unsafe recording system may result in electric shocks and other injuries of the subject or other persons.

Sensors may be equipped with two or three output leads terminated with 1.5-mm touchproof safety connectors which fit into the amplifier's input connectors. Please connect the sensors as follows, depending on the sensors and amplifiers.

1.3.1 g.USBAMP

Sensors with two leads:

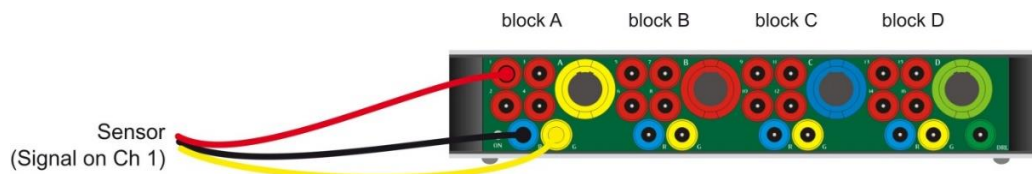
Connect one lead (black, if applicable) to the REF input of g.USBamp (blue) and the other lead (red, if applicable) to an input channel of g.USBamp (red). Use a jumper cable to shortcut the REF and GND of the used block of the amplifier.



Sensors with three leads:

Plug the yellow connector into the GND socket (yellow). Plug the blue (or black) connector into the REF socket (blue). Plug the red connector into an input channel socket of g.USBamp (red). In this case, the sensor output cable shorts REF and GND of the used input block of g.USBamp.

Note: Do not use other input channels of the used block for electrophysiological recordings (EEG, ECoG, EMG, ECG, EOG, etc.) because the shortcut between GND and REF may reduce the quality of such signals. Do not use the "Common REF" functions for this block to avoid interference with other input blocks.



1.3.2 g.HIAMP

Use a 64-channel passive electrode connector box or a 16-channel passive electrode connector box for g.HIamp.

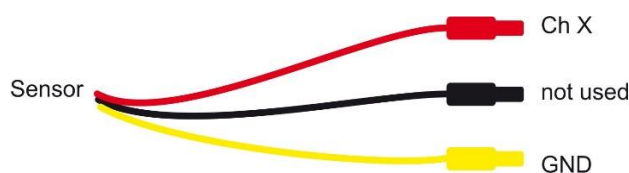
Sensors with two leads:

Connect one lead (black, if applicable) to the GND socket (yellow) and connect the other lead (red, if applicable) to an input channel socket (red).



Sensors with three leads:

Plug the yellow connector into the GND socket (yellow) and plug the red connector into an input channel socket (red). The blue (or black) connector remains unused in this case.



1.3.3 g.NAUTILUS (MULTI-PURPOSE)

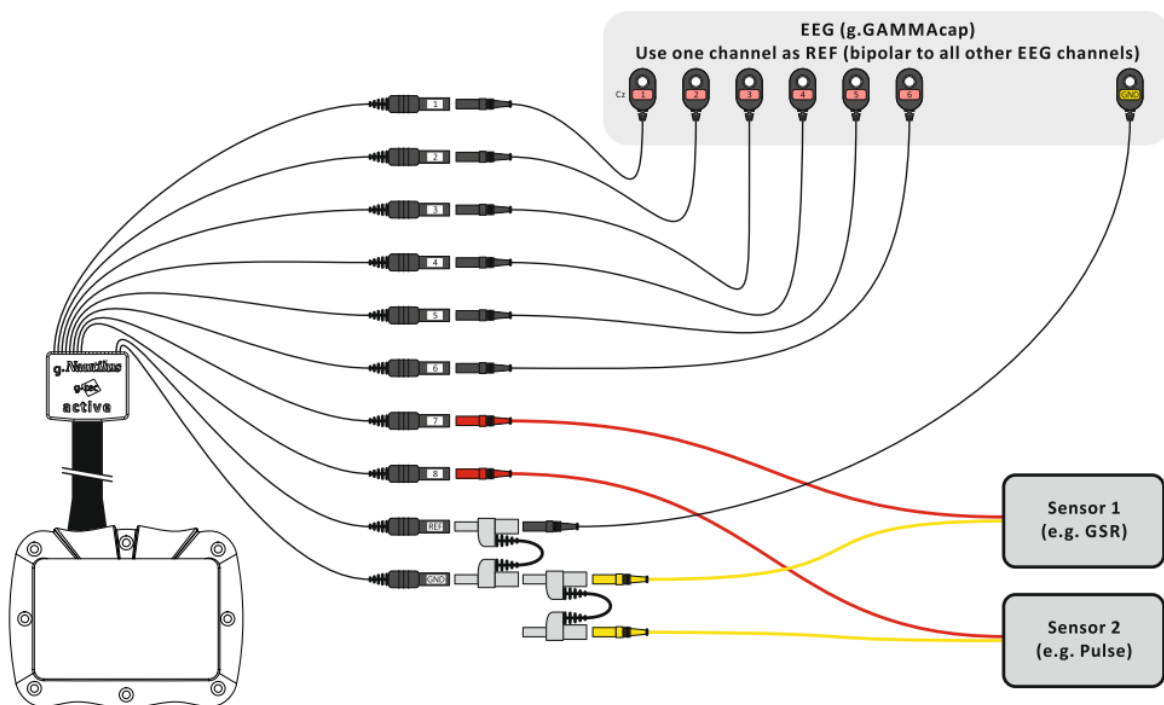
Disconnect the REF and the GND electrodes from the device and use a shortcut jumper cable to shortcut REF and GND. The GND electrode needs to be connected to the jumper cable again. For using multiple sensors use multiple jumper cables.

Please note that for most sensors the input voltage range of g.Nautilus needs to be increased to +/- 375 mV to avoid channel saturation!

Sensors with three leads:

The black connector of the sensor remains unused.

Sensors with two leads:



2 g.SENSORS

2.1 TEMPERATURE



What can be measured?

The g.SENSOR Temperature is designed to measure the skin temperature within a range of 20 – 45 °C. The skin temperature depends on skin perfusion, which is influenced by physiological and psychological parameters.

How does the sensor work?

The sensor is equipped with a high-accuracy YSI-400 thermo sensor to be applied directly on the skin. Changes of the skin temperature are transformed into a proportional voltage output signal. The sensor is already calibrated, and 0 V output voltage corresponds to 20°C. A temperature increase of 1°C adds 8 mV to the output signal, so that 45°C will yield an output voltage of 200 mV.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output cable and requires a DC-coupled amplifier (no high pass filter!), lowpass filter 30 Hz, notch filter 50/60 Hz on.

Correct application and testing

Place the sensor on the skin (e.g. on the finger, forehead or cheek) and fix it with tape or with an elastic band. Make sure that the sensor sits smoothly on the skin with little pressure. Note that high accuracy skin temperature recording requires a room with constant temperature and humidity, and the subject needs to adapt to these conditions for 30 – 60 minutes prior to the experiment. Make sure that the interface box is turned on, and replace the battery if the green indicator LED is not brightly lit.

Safety measures

Use 9V alkaline batteries or rechargeable batteries only. Do not connect the sensor to any power supply or charging device!

Technical data and specifications

Power supply: 9V alkaline battery or rechargeable battery

Current consumption: < 10 mA

Output signal range: 0 – 200 mV (calibrated for 20 – 45°C)

Accuracy: 0.2°C

g.SENSOR Temperature is not a certified medical device. The product is not intended to be used as a medical device.



NOTE

Using this sensor with g.Nautilus Multi Purpose requires an increase of the input voltage range to +/- 375 mV!

2.2 GALVANIC SKIN RESPONSE



What can be measured?

The g.SENSOR Galvanic Skin Response measures the electrical conductivity of the skin, also known as galvanic skin response (GSR) or electro-dermal activity (EDA), which depends on perspiration. Psychological and physiological stress and emotional responses activate sweat glands, thereby increasing perspiration and skin conductivity. The signal can change quickly, such as in response to stimulation, or slowly, such as an adaptation to the environment or to a general stress level.

How does the sensor work?

A small constant DC voltage is applied to a pair of special finger electrodes and the current is measured by the sensor. The output signal is proportional to the skin conductivity in a range of 0 – 30 $\mu\text{S}/\mu\text{Mho}$ (micro Siemens/micro Mho). The sensor is equipped with an internal isolation barrier to rule out influences between GSR electrodes and other electrophysiological recordings.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output cable and requires a DC-coupled amplifier to record conductivity levels and slow changes (no high pass filter!), lowpass filter 30 Hz, and notch filter at 50 or 60 Hz. To record only quick responses, an additional high pass filter (e.g. 0.05 or 0.1 Hz) can be used.

Correct application and testing

Place the electrodes on two fingers of the same hand (electrode contacts on the opposite side of the fingernail) and close the Velcro straps with a little tension. Turn on the device and replace the battery if the green indicator LED is not brightly lit. Watch the signal and wait for a few minutes till the electrodes have adapted. Short hyperventilation (heavy breathing) will cause an immediate increase of skin conductivity in most subjects. Note that GSR signals can be very individual. Some subjects show high fluctuations and responses even without any obvious stimulus, while others will show only very small changes over a long time. Pressing the calibration button adds +1 μS to the current value and results in a +6.7 mV signal step for testing purposes.

Safety measures

Use 9V alkaline batteries or rechargeable batteries only. Do not connect the sensor to any power supply or charging device!

Technical data and specifications

Power supply: 9V alkaline battery or rechargeable battery

Current consumption: < 40 mA

Output signal range: 0 – 200 mV (corresponding to 0 – 30 $\mu\text{S}/\mu\text{Mho}$)

Electrode voltage: approx. 400 mV DC

Isolation barrier: 500 V RMS between electrodes and output signal

g. SENSOR Galvanic Skin Response is not a certified medical device. The product is not intended to be used as a medical device.



NOTE

Using this sensor with g.Nautilus Multi Purpose requires an increase of the input voltage range to +/- 375 mV!

2.3 ACCELERATION



What can be measured?

The g.SENSOR Acceleration measures G-forces in 3 directions (X/Y/Z) in a range of ± 3 g. It can be used to record acceleration or vibration values of up to 80 Hz. When attached to the subject's body, the sensor measures changes of position or physical activity (such as walking, running or jumping). The g.Gsensor can also measure acceleration and vibration when mounted on an object (vehicle, aircraft, simulator, etc.)

How does the sensor work?

The internal sensing element measures G-forces of the orthogonal X-, Y- and Z-axes independently. For each axis, an output signal is provided, and $+1$ g equals an output voltage of $+66$ mV.

Recommended amplifier configuration

The sensor signals are measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output cable and requires a DC-coupled amplifier (no high pass filter!), lowpass filter 30 – 100 Hz, and notch filter at 50 or 60 Hz. To record faster movements such as vibration, the filters may be set according to the expected activity and may include a high pass filter.

Correct application and testing

Attached to the body: Use the belt clip or other suitable materials to position the sensor on an appropriate location.

Fixed to an object: Make sure that the sensor is securely mounted and cannot move.

Shake the sensor and turn it around the X-, Y- and Z-axes to check all signals. The directions of the axes are indicated on the device. Replace the battery if the green indicator LED is not brightly lit.

Safety measures

Use 9V alkaline batteries or rechargeable batteries only. Do not connect the sensor to any power supply or charging device!

Technical data and specifications

Power supply: 9V alkaline battery or rechargeable battery

Current consumption: < 10 mA

Output signal range: -200 to +200 mV (equals -3 g to +3g)

Shock resistance: max. 10 g

Accuracy: 5%

Nonlinearity: <0.3 %

g. SENSOR Acceleration is not a certified medical device. The product is not intended to be used as a medical device.



NOTE

Using this sensor with g.Nautilus Multi Purpose requires an increase of the input voltage range to +/- 375 mV!

2.4 RESPIRATION AIRFLOW



What can be measured?

The Respiration Airflow Sensor measures breathing activity through mouth and nose. The output signal voltage rises during inhalation and drops during exhalation.

How does the sensor work?

The sensor measures the air temperature right in front of the nose and mouth. As exhaled air is normally warmer than the air in the room, the temperature changes reflect respiratory activity.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output and requires a bandpass filter 0.1 Hz – 30 Hz and notch filter 50/60 Hz on

Correct application and testing

The sensor is placed on the upper lip, below the subject's nostrils. Bend the two nasal prongs forward to prevent them from entering the nostrils. Put the sensor wires on each side of the sensor over the patient's ears and secure them with tape. Then, lay the wires under the chin and close the lock. Connect the sensor to the interface box and check the signal.

Safety measures

Please refer to the manufacturer's instructions for use, which are packed together with the sensor.

Technical data and specifications

Power supply: internal battery in the interface box with a minimum life time of 1 year. Replace the interface box when the battery is empty and the sensor stops working.

Sensor type: Thermistor

Output signal range: +/- 0.5 to 1 mV (typical)

The Respiration Airflow Sensor is not a certified medical device. The product is not intended to be used as a medical device. Please also read the manufacturer's instructions for use, which are packed with the sensor.

2.5 RESPIRATION EFFORT



What can be measured?

The Respiration Effort Sensor measures changes of the circumference of the torso related to breathing activity. It can be applied around the chest or abdomen, or on both of these areas to compare abdominal and chest respiratory activity.

How does the sensor work?

The sensor measures the tension of an elastic band using a piezo-electric element. Changes of the tension cause a voltage output indicating changes of the torso's circumference.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output and requires a bandpass filter 0.1 Hz – 30 Hz and notch filter 50/60 Hz on

Correct application and testing

The sensor element is connected to the elastic band with two snap locks. The length of the band should be adjusted to provide some tension when attached around the torso. Place the sensor element at a position where it will not be touched accidentally to avoid artifacts. Movements of the sensor element and the cables may cause large artifacts in the signal.

Safety measures

Please refer to the manufacturer's instructions for use, which are packaged together with the sensor.

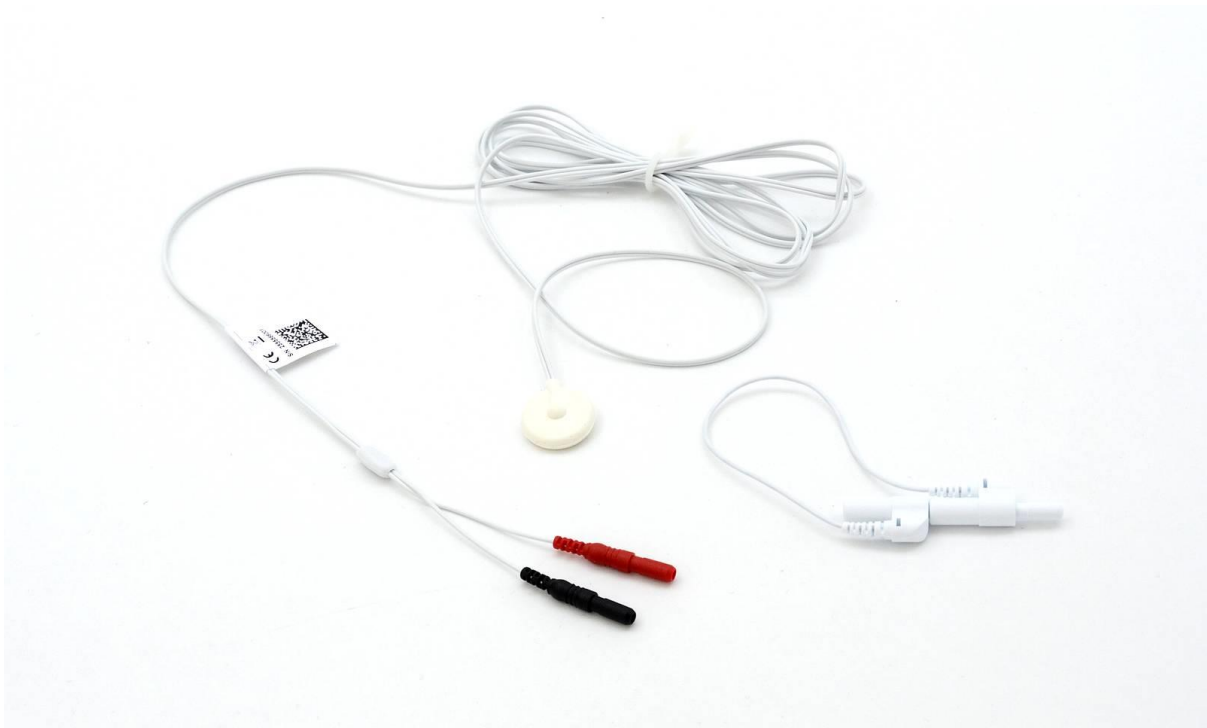
Technical data and specifications

Sensor type: Piezo-electric tension sensor

Output signal range: +/- 1 mV (typical)

The Respiration Effort Sensor is not a certified medical device. The product is not intended to be used as a medical device. Please also read the manufacturer's instructions for use, which are packaged together with the sensor.

2.6 SNORING



What can be measured?

The snoring sensor records tracheal sound and is predominantly used in sleep research to detect snoring in sleeping subjects.

How does the sensor work?

The sensor uses a piezo-electric element which picks up vibrations and low frequency sound from the skin surface.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output and requires a bandpass filter 5 Hz – 100 Hz and notch filter 50/60 Hz on

Correct application and testing

The sensor is attached to the skin with adhesive tape. Place the sensor on the subject's neck close to the trachea or larynx. For sleep research make sure that the wires can move freely with the subject.

Safety measures

Please refer to the manufacturer's instructions for use, which are packaged together with the sensor.

Technical data and specifications

Sensor type: Piezo-electric sensor

Output signal range: +/- 0.5 to 5 mV (typical)

The snoring sensor is not a certified medical device. The product is not intended to be used as a medical device. Please also read the manufacturer's instructions for use, which are packaged together with the sensor.

2.7 LIMB MOVEMENT



What can be measured?

The limb movement sensor is used to identify sudden movements of the legs or hands during sleep (e.g. in patients suffering from restless legs syndrome). The sensor provides a clear signal indicating each movement, which allows investigation of the frequency of movements during the night.

How does the sensor work?

The sensor uses a piezo-electric element, which is attached to an elastic Velcro holster band. The sensor is fixed at the ankle or wrist and generates a voltage output signal corresponding to movement related changes of tension of the holster band.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output and requires a bandpass filter 0.1 Hz – 60 Hz and notch filter 50/60 Hz on

Correct application and testing

Apply the sensor with the elastic Velcro holster band at the ankle or wrist. Close the band with a little tension and test the signal output. Please also see the manufacturer's instructions in the manual.

Safety measures

Please refer to the manufacturer's instructions for use, which are packaged together with the sensor.

Technical data and specifications

Sensor type: Piezo-electric sensor

Output signal range: +/- 0.5 to 5 mV (typical)

The limb movement sensor is not a certified medical device. The product is not intended to be used as a medical device. Please also read the manufacturer's instructions for use, which are packaged together with the sensor.

2.8 OXYGEN SATURATION



What can be measured?

The oxygen saturation sensor measures the oxygen saturation of the peripheral blood in percent. The pulse frequency (0 – 300 bpm) and the plethysmogram (auto scaled) can also be recorded.

How does the sensor work?

The sensor sends infrared light (LED) into the tissue and measures the amount of reflected light at different wavelengths. Since the color of blood depends on the amount of solute oxygen, the percentage concentration (SpO2) can be computed. All signals are delivered as analog voltage signals.

Recommended amplifier configuration

The sensor signals are measured by gtec's biosignal amplifiers via touch proof connectors of the sensor output cable and require following settings:

- SpO2-signal (pin 1, red connector): DC required! (0 Hz), Lowpass: 30 Hz, notch filter 50/60 Hz on
- Pulse frequency (pin 2, black connector): DC required! (0 Hz), Lowpass: 30 Hz, notch filter 50/60 Hz on
- Plethysmogram (pin 3, white connector): 0.1 – 30 Hz, notch filter 50/60 Hz on

Correct application and testing

The finger sensor should be placed on the index finger of the right or left hand with the LED on the opposite side of the fingernail. Turn on the device by pressing the "ON/OFF" button. Immediately after turning on the device, a calibration signal will be delivered on the SpO2 and the pulse frequency outputs for a few seconds. These rectangular signals alternate between 0% and 100% (SpO2) and between 0 bpm and 300 bpm (Pulse frequency), respectively. After some seconds, the "SIG OK" indicator should turn on permanently. The LED for "BAT OK" indicates proper battery voltage.

Safety measures

Use 2 x 1.5V alkaline batteries (AA) only. Do not connect the sensor to any power supply or charging device!

Technical data and specifications

Output signal range: approx. 0 – 200 mV

Power supply: 2 x 1.5 V (AA) alkaline batteries, internal fuse: 500 mA

Current consumption: < 60 mA

BAT OK voltage: >2.4 V SIG OK: quality > 90 %

The oxygen saturation sensor is not a certified medical device. The product is not intended to be used as a medical device.



NOTE

Using this sensor with g.Nautilus Multi Purpose requires an increase of the input voltage range to +/- 375 mV!

2.9 BLOOD PRESSURE



What can be measured?

The CNAP-AUX Analog Out Adapter is a signal interface box to provide an analog voltage signal of the Continuous Non-invasive Arterial blood Pressure and other parameters measured by the CNAP monitor (CNSystems Medizintechnik AG, Austria). The following parameters can be measured depending on the version of the CNAP monitor via analog output channels CH1 to CH4 (see table below). For CH2 to CH4 different parameters can be selected in the setup menu of the CNAP monitor. Please see the manual of the CNAP monitor for details and descriptions of the parameters and abbreviations.

CNAP		Monitor Software Version			
Adapter output		V5.0.x	V5.2.0 to 5.2.14 ¹		V5.2.16 and later ¹
CH1	BP waveform 0 - 500 mmHg		BP waveform 0 - 500 mmHg		BP waveform 0 - 500 mmHg
CH2	MAP 0 - 500 mmHg	selectable	MAP ² 0 - 300 mmHg Pulse 0 - 200 bpm	selectable	MAP ² 0 - 300 mmHg Pulse 0 - 200 bpm
CH3	CO 0 - 100 l/min	selectable	CO ² 0 - 30 l/min SV 0 - 200 ml SVR 0 - 5000 dyn*s/cm ⁵	selectable	sysBP 0 - 300 mm/Hg CO ² 0 - 30 l/min SV 0 - 200 ml SVR 0 - 5000 dyn*s/cm ⁵
CH4	PPV 0 - 100%	selectable	PPV 0 - 40% SVV 0 - 40% Pulse ² 0 - 200 bpm	selectable	diaBP 0 - 300 mm/Hg PPV 0 - 40% SVV ³ 0 - 40% Pulse ² 0 - 200 bpm

¹ Since version 5.2.0 the parameter output on channel 2, 3 and 4 can be configured by the user in the menu **Setup | Measurement | Output Options**, ²Default setting, ³ The parameter SVV is not available on the US market

How does the sensor work?

The CNAP monitor measures the continuous arterial pressure and other parameters by using two finger cuffs and one arm cuff for calibration. Four analog output channels are provided at the AUX Analog Output socket of the device. These signals are isolated from mains and are safe to be recorded via the input channels of g.tec amplifiers. The CNAP AUX Analog Out Adapter is needed to reduce the output voltage level from 5 V to 200 mV or from +/- 5 V to +/- 200 mV depending on the selected setup of the CNAP monitor. The signal ranges given in the table above correspond to these voltage ranges recorded with g.tec amplifiers.

Recommended amplifier configuration

The sensor signal is measured by gtec's biosignal amplifiers via touch proof connectors of the adapter and requires DC-coupled inputs, low-pass filter should be set to 30 Hz – 60 Hz, notch filter 50/60 Hz on.



NOTE

Using this sensor with g.Nautilus Multi Purpose requires an increase of the input voltage range to +/- 375 mV!

Correct application and testing

Carefully read and follow the manufacturer's manuals of the CNAP monitor (delivered with the device) before using the device. Plug the "CNAP AUX" plug to the "CNAP AUX" socket on the right side of the CNAP monitor. Plug the GND wire (yellow) to the GND socket of g.Hlamp or to the short-cut REF and GND sockets of g.USBamp or g.Nautilus. Plug the CH1 – CH4 wires (red) to four free input channels of the amplifier and check the signals.



NOTE

The analog output signals are only provided after the CNAP monitor has finished the initialization and calibration procedure, which takes several minutes



NOTE

The g.SENSOR Blood Pressure is not a certified medical device. The product is not intended to be used as a medical device.

3 TRIGGER

3.1 G.TRIGBOX



What is g.TRIGbox?

The g.TRIGbox is a tool to generate precise bounce-free trigger pulses from various sensor or digital input signals. Input and output lines are isolated from each other. The trigger outputs can be connected to a data acquisition system, to the trigger inputs or to the analog input channels of biosignal amplifiers. Output trigger pulses have a length of at least 20 ms to ensure secure trigger detection of the recording system, even at low sampling frequencies. Four trigger outputs are available, and each output can be controlled via three different types of input signals (A, B, C, D):

- 1) Low level input signals in the range of ± 0.5 mV to ± 200 mV via dynamic microphones or other sensors (e.g. inductive sensors, piezo sensors, sound card output signals, ...)
- 2) High level input signals in the range of ± 100 mV to ± 5 V via push buttons or other input devices (e.g. digital outputs from various devices with CMOS or TTL logic levels)
- 3) Optical sensor signals via highspeed phototransistor sensors to detect changes of light intensity e.g. on a computer screen, on LEDs, flash lamps and other light sources

Each channel has a separate threshold level adjustment and a trigger output indicator LED. The trigger output signals are provided at two different voltage levels: 5 V (for TTL or CMOS logic inputs) and 200 mV (for the connection to analog unfiltered amplifier inputs). In addition to the single trigger output signals, an encoded 4-bit analog output signal can be used to record all 16 possible output combinations in one analog channel.

Correct application and testing

Connect a trigger source to an appropriate input channel of the g.TRIGbox and turn on the device. The LOW BATT indicator LED will flash for about half a second. If the LED remains on or does not flash

at all, the battery needs to be replaced. Then, activate the trigger source and adjust the threshold for reliable and secure trigger detection. The trigger output LED indicates a HI output signal. When using analog trigger sources or optical sensors, the accurate adjustment of the threshold is very important. Identify the lower detection threshold (output trigger remains LO) and the upper detection threshold (output trigger always HI) by turning the adjustment knob counter-clockwise and clockwise, respectively. Remember the knob positions and adjust the knob then about to the middle in between these two positions. Connect the trigger output cable to the input channel of the amplifier and check the trigger signal or marker.

Safety measures

Use 9-V alkaline batteries or the optional safety power supply only. Do not connect g.TRIGbox to any charging devices or 3rd-party power supplies.

Technical data and specifications

Power supply: 9V alkaline battery or optional safety power supply.

Current consumption: < 60 mA

Isolation voltage between inputs and outputs: 4 kV



NOTE

g.TRIGbox is not a certified medical device. The product is not intended to be used as a medical device.

connect optical sensors to detect screen markers
flash lamps, LED indicators, slide projectors etc.

connect high level trigger sources like
push buttons, TTL signals, or other
logical signals

adjust threshold, set threshold level
about the middle between the upper
and lower limit



optionally connect -when necessary
e.g. for low level trigger sources- the
laboratory ground to the input GND
socket via the green grounding cable

connect low level trigger sources
such as microphones, sound card,
piezo sensors, inductive sensors

low battery indicator:
Short flash when switching -> OK
permanent on -> replace battery

switch ON/OFF

external safety power supply

trigger output indicators

g.TRIGbox Output Function	15-Pin D-SUB Socket
GND	Pin 1
Output A (5V)	Pin 2
Output B (5V)	Pin 3
Output C (5V)	Pin 4
Output D (5V)	Pin 5
Encoded analog output (5V)*	Pin 8
Output A (200 mV)	Pin 9
Output B (200 mV)	Pin 10
Output C (200 mV)	Pin 11
Output D (200 mV)	Pin 12
Encoded analog output (200 mV)*	Pin 15

*Encoded analog output:

Output A – least significant bit

Output D – most significant bit

Output voltage steps: 12.5 mV for 200 mV BSL

3.2 ADAPTER FOR TRIGGER CABLE G.TRIGBOX FOR G.USBAMP/G.HIAMP/G.NAUTILUS

This cable connects the output socket of g.TRIGbox to a 25-Pin D-SUB female connector.

3.2.1 PIN-ASSIGNMENT

g.TRIGbox Output Function 15-Pin D-SUB Connector		25-Pin D-SUB Connector
Trigger output A	Pin 2	Pin 2
Trigger output B	Pin 3	Pin 3
Trigger output C	Pin 4	Pin 4
Trigger output D	Pin 5	Pin 5
Digital GND	Pin 1	Pin 18-25

3.3 USB-TO-PARALLEL PORT ADAPTER



What is USB-to-Parallel-Port Adapter?

The USB-to-Parallel Port Adapter sends 8-bit trigger information via software through a USB to serial port connection to the digital inputs of g.USBamp, g.Nautilus, and g.Hlamp (see [Chapter 2 Trigger cables to 25-PIN D-SUB male CONNECTOR](#) for details). A total of 255 different trigger codes can be encoded and transmitted.

Sending triggers via the serial port is available in major stimuli presentation packages, among them

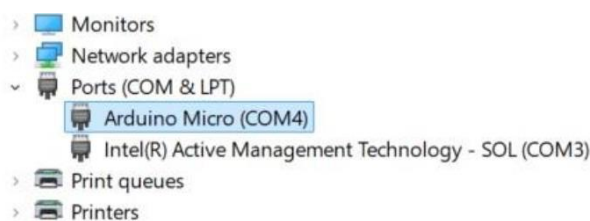
Presentation from Neurobehavioral Systems, Inc.,
E-Prime from Psychology Software Tools,
PsychoPy from the University of Nottingham,
MATLAB from MathWorks, Inc.,
and any other stimuli software that send a trigger code (event marker) via serial port communication.

The device is based on Arduino Micro circuit which has a built-in USB communication. This yields a reduced trigger delay, which is typically in the one millisecond range. An adapter control software is already installed on the device.

Correct installation and testing

For Windows 10 or higher install the Arduino IDE software which is available via www.arduino.cc. Then connect the USB-to-Parallel Adapter and the device driver will be installed.

By running the "Device manager" take a note of the COM port used e.g. "Arduino Micro (COM4)"



In this case, the COM port is "COM4".

By default, the serial port properties are the following:

Baud rate: 9600

Data bits: 8

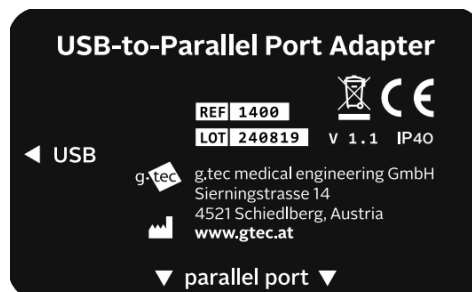
Parity: none

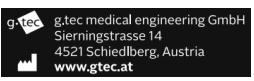
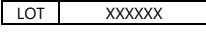
Stop bit: 1

Code example in Python

```
import serial          #Import the serial library
val = 10               #trigger
val_inbytes= val.to_bytes(8, byteorder='big')
s_port = serial.Serial()
s_port.port = 'COM4'
s_port.baurate=9600
s_port.bytesize=8
s_port.parity='N'
s_port.stopbits=1
s_port.timeout = 1
s_port.open()
s_port.write(val_inbytes)
val = 0
val_inbytes= val.to_bytes(8, byteorder='big')
s_port.write(val_inbytes)
s_port.close()
```

USB-to-Parallel port Adapter – label



Label	Description
	Marks the position of the socket for the micro USB cable.
	Marks the position of the parallel port socket.
	Manufacturer address.
	Lot number of the device

REF	1400
-----	------

Order number of the device.

V X.X

Device Version



Do not dispose the device with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.



CE mark

IP40

Protection against solid objects > 1 mm

Technical data and specifications

Power supply: 5 V DC via USB.

Rated power consumption: ≤ 1 W

Digital trigger output: 5 V DC



NOTE

USB-to-Parallel port adapter is not a certified medical device. The product is not intended to be used as an accessory for a medical device.

3.4 G.AUDIOBOX



What is g.AUDIObox?

The g.AUDIObox plays stereo sound via the connected headphones. An audio extension cable is included. A digital output is set when the audio file starts playing. This trigger information is sent to the connected digital input of a g.tec amplifier. Data is transferred from the control PC to the g.AUDIObox via a USB-C cable. On command the sound can be played.

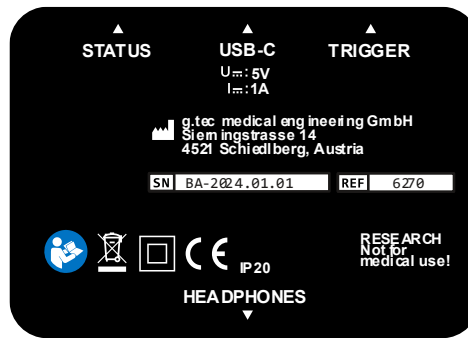


Application

Install and setup the corresponding software for the device. Connect the USB cable, the g.AUDIObox headphones and the trigger cable to the g.AUDIObox. Now connect the USB cable to the computer and the cyan STATUS LED lights up. Connect the trigger cable to the digital input cable of your g.tec biosignal amplifier. For more information about the trigger cable, see 3.6. Further details on how to install and use the g.AUDIObox and application examples are described in the Instruction for Use of g.Hlsys.



g.AUDIObox - label



Label	Description		
 STATUS	Marks the position of the status LED on the front side.		
 USB-C	Marks the position of the socket for the USB-C cable.		
 TRIGGER	Marks the position of the socket for the trigger cable.		
 g.tec medical engineering GmbH Sierningstrasse 14 4521 Schiedlberg, Austria	Manufacturer address.		
<table border="1"><tr><td>SN</td><td>BA-2024.01.01</td></tr></table>	SN	BA-2024.01.01	Serial number in the format BA-Year_of_production.Month.Number
SN	BA-2024.01.01		
<table border="1"><tr><td>REF</td><td>6270</td></tr></table>	REF	6270	Order number of the device.
REF	6270		
RESEARCH Not for medical use!	The device is intended to be used in research and scientific investigations. The devices must not be used for medical applications nor for treatment or diagnosis in medical environments.		
	Follow instructions for use.		
	Do not dispose the device with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.		



Safety class II



CE mark

IP20

Protection against solid objects > 12.5 mm



Marks the position of the socket for the headphones.

Technical data and specifications

Power supply: 5 V DC via USB.

Rated power consumption: ≤ 5 W

Digital trigger output: 3.3 V DC

Digital trigger length: 20 ms

Audio latency (software command to audio output): < 200 ms

Audio output to digital trigger latency: < 10 ms



NOTE

g.AUDIObox is not a certified medical device. The product is not intended to be used as a medical device.



WARNING

Listening at high volume for long periods may damage your hearing.

3.4.1 G.AUDIOBOX HEADPHONES

The g.AUDIObox includes on-ear headphones (JBL TUNE 500, 32-ohm). Using other models may cause improper audio levels.

3.5 g.TACTILEbox



What is g.TACTILEbox?

The g.TACTILEbox delivers vibro-tactile stimulation via seven connected stimulators. A digital output is set when a stimulation is switched on. This trigger information is sent to the connected digital input of a g.tec amplifier. Data is transferred from the control PC to the g.TACTILEbox via a USB-C cable. On command the vibration of each stimulator can be switched on or off. Up to three stimulators can be switched on simultaneously.

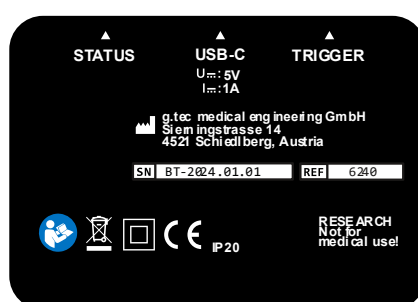


Application

Install and setup the corresponding software for the device. Connect the USB cable and the trigger cable to the g.TACTILEbox. Now connect the USB cable to the computer and the cyan STATUS LED lights up. Connect the trigger cable to the digital input cable of your g.tec biosignal amplifier. For more information about the trigger cable, see 3.6. Further details on how to install and use the g.TACTILEbox and application examples are described in the Instruction for Use of g.Hlsys.



g.TACTILEbox – label



Label	Description
	Marks the position of the status LED on the front side.
	Marks the position of the socket for the USB-C cable.
	Marks the position of the socket for the trigger cable.
	Manufacturer address.
	Serial number in the format BA-Year_of_production.Month.Number
	Order number of the device.
	The device is intended to be used in research and scientific investigations. The devices must not be used for medical applications nor for treatment or diagnosis in medical environments.



Follow instructions for use.



Do not dispose the device with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.



Safety class II



CE mark

IP20

Protection against solid objects > 12.5 mm

Technical data and specifications

Power supply: 5 V DC via USB.

Rated power consumption: ≤ 5 W

Digital trigger output: 3.3 V DC

Digital trigger length: 20 ms

Stimulation output to digital trigger latency: < 10 ms

Stimulation latency (software command to stimulation onset): < 50 ms

Stimulation frequency: 80 Hz – 150 Hz (depending on the fixation and montage)



NOTE

g.TACTILEbox is not a certified medical device. The product is not intended to be used as a medical device.

3.6 ADAPTER FOR TRIGGER CABLE G.AUDIOBOX/G.TACTILEBOX FOR G.USBAMP/G.HIAMP/G.NAUTILUS

The adapter for single trigger connects the output socket of g.AUDIObox or g.TACTILEbox to the 25-Pin D-SUB female connector. The adapter for double trigger connects the output of g.AUDIObox and the output of g.TACTILEbox to the digital input of g.Nautilus.

Adapter for single trigger for g.AUDIObox/
g.TACTILEbox.



Adapter for double trigger for g.AUDIObox
and g.TACTILEbox.



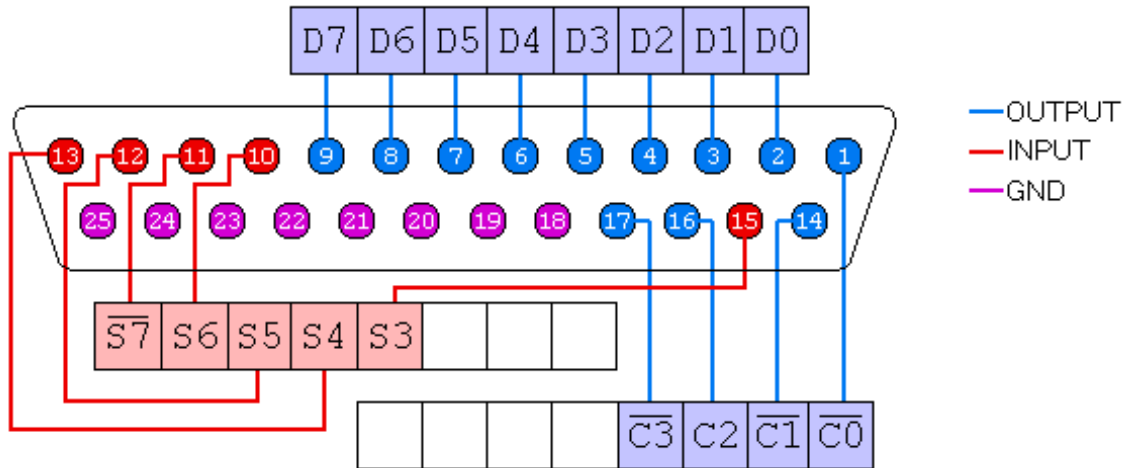
3.6.1 PIN-ASSIGNMENT

g.AUDIObox/ g.TACTILEbox Output Function	3-Pin 2.5 mm headphone plug	25-Pin D-SUB Connector	26 pin D-SUB Connector g.Nautilus
Trigger output g.AUDIObox	Pin A (left audio channel)	Pin 2	Pin 1
Trigger output g.TACTILEbox	Pin B (right audio channel)	Pin 3	Pin 2
Digital GND	Pin C (GND)	Pin 18-25	Pin 19

4 TRIGGER CABLES TO 25-PIN D-SUB MALE CONNECTOR

To use 3rd-party stimulation software such as E-Prime or Presentation, trigger cables with 25-pin D-SUB connectors are provided to be plugged into the PC's parallel port socket or corresponding compatible devices.

4.1 PIN-ASSIGNMENT (PARALLEL PORT)



4.2 TRIGGER CABLE g.USBAMP TO 25-PIN D-SUB CONNECTOR

Plug the connectors of the Digital input cable to DIG I/O 1 and DIG I/O 2 into the corresponding sockets at the rear side of the amplifier.

4.2.1 PIN-ASSIGNMENT AND DIGITAL CHANNELS

D-SUB 25 (Plug, Male Pins)	D-Sub 25 Function	DIG I/O Function
1		
2	D0	Digital input 1
3	D1	Digital input 2
4	D2	Digital input 3
5	D3	Digital input 4
6	D4	Digital input 5
7	D5	Digital input 6
8	D6	Digital input 7

9	D7	Digital input 8
10	S6	Digital output 1
11		
12	S5	Digital output 2
13	S4	Digital output 3
14		
15	S3	Digital output 4
16		
17		
18		
19		
20		
21		
22		
23		
24	GND	
25	GND	

4.3 TRIGGER CABLE g.HIAMP TO 25-PIN D-SUB MALE CONNECTOR

Plug the connector of the Digital input cable into the DIG IN 1 or into the DIG IN 2 sockets. DIG IN 1 supports digital input channels 1 to 8 and DIG IN 2 supports digital input channels 9 to 16.

4.3.1 PIN-ASSIGNMENT AND DIGITAL CHANNELS DIG IN1

D-SUB 25 (Plug, Male Pins)	D-Sub 25 Function	DIG IN 1 Function
1		
2	D0	Digital input 1
3	D1	Digital input 2

4	D2	Digital input 3
5	D3	Digital input 4
6	D4	Digital input 5
7	D5	Digital input 6
8	D6	Digital input 7
9	D7	Digital input 8
10		
11		
12		
13		
14		
15		
16		
17		
18	GND	
19	GND	
20	GND	
21	GND	
22	GND	
23	GND	
24	GND	
25	GND	

4.3.2 PIN-ASSIGNMENT AND DIGITAL CHANNELS DIG IN2

D-SUB 25 (Plug, Male Pins)	D-SUB 25 Function	DIG IN 2 Function
----------------------------	-------------------	-------------------

1

2	D0	Digital input 9
3	D1	Digital input 10
4	D2	Digital input 11
5	D3	Digital input 12
6	D4	Digital input 13
7	D5	Digital input 14
8	D6	Digital input 15
9	D7	Digital input 16
10		
11		
12		
13		
14		
15		
16		
17		
18	GND	
19	GND	
20	GND	
21	GND	
22	GND	
23	GND	
24	GND	
25	GND	

4.4 TRIGGER CABLE FOR g.NAUTILUS TO 25-PIN D-SUB MALE CONNECTOR

Plug the connector of the trigger cable into the digital input socket of the g.Nautilus Base station.

4.4.1 PIN-ASSIGNMENT AND DIGITAL CHANNELS

D-SUB 25 (Plug, Male Pins)	D-SUB 25 Function	DIG IN Function
1		
2	D0	Digital input 1
3	D1	Digital input 2
4	D2	Digital input 3
5	D3	Digital input 4
6	D4	Digital input 5
7	D5	Digital input 6
8	D6	Digital input 7
9	D7	Digital input 8
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		

23

24

GND

25

5 USB GROUNDING CABLE

Biopotential recordings in general are quite sensitive to several disturbing influences such as movements of the subject, the cables and the electrodes or coupling of electromagnetic fields and transient electrical events like electrostatic discharges in the vicinity of the recording. Depending on your current laboratory layout, laboratory grounding concept, and/or the used power supply of the laboratory recording PC, a 50 Hz (60 Hz) mains power supply humming noise might interfere the EEG measurements when using g.Hlamp. This phenomenon is called common mode interference. In such situations the laboratory grounding cable might help reducing these interferences.



WARNING

Make sure that the laboratory grounding complies with the local standards for electrical installations.

Connect the USB connector (PC USB) to a free USB socket on your PC. Connect the safety alligator clip (Lab GND) to your laboratory grounding point.



PC USB

Lab GND



laboratory grounding point

5.1 PIN-ASSIGNMENT

USB-A (Plug, Pins)	Function	Alligator clip
1	Not connected	
2	Not connected	
3	Not connected	
4	GND	GND

6 CONNECTING g.tec's AMPLIFIERS TO E-PRIME

In E-Prime, the parallel port can be used with the Basic programmable InLine object. In MATLAB/Simulink, the signal can be recorded with the Simulink Highspeed model "DigitalInputDemo.sltx". This model is part of the g.tec Highspeed On-Line Processing for Simulink package.

Please refer to the g.Hlsys user manual

g·SENSOR
BIOSIGNAL SENSOR